

CSCE 46103/56103 Artificial Intelligence – Fall 2026

MDP Formulation

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Sequential Decision Problems

A Single Decision

The game-show choice: take \$1 million, or a fair coin toss paying \$0 or \$2.5 million.

$$d^*(e) \in \arg \max_d EU(d \mid e)$$

Compare expected utilities using the available information e .

Sequential Decision Making

Which action should an agent choose when its consequences are uncertain?

- ▶ Actions change future situations.
- ▶ Rewards express what matters.
- ▶ Later choices depend on what happens.

Sequential Decision Problems

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Today's Model

A Markov decision process (MDP) describes a fully observed state, uncertain transitions, and rewards over time.

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A Robot in a Stochastic Grid

3				+1
2		wall		-1
1	start			
	1	2	3	4

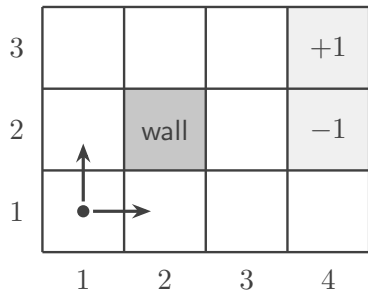
- ▶ The robot observes its current cell.
- ▶ It chooses up, right, down, or left.
- ▶ A move can slip sideways.
- ▶ The wall cannot be entered.
- ▶ Entering either terminal cell ends the task.

Coordinates are (x, y) : column first, then row.
The start is $(1, 1)$.

Decision Question

How should the robot act to reach a rewarding outcome while accounting for possible slips?

Action Outcomes at a Boundary

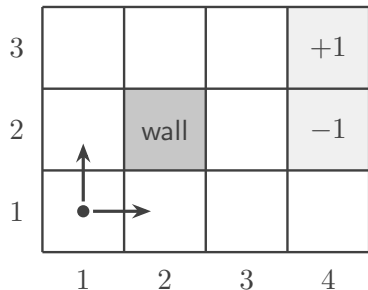


Selected action: up at (1, 1).

Attempt	Next cell	Probability
Up	(1, 2)	0.8
Right	(2, 1)	0.1
Left (blocked)	(1, 1)	0.1
Down (blocked)	(1, 1)	0

Downward and leftward attempts hit the boundary, leaving the robot at (1, 1).

Action Outcomes at a Boundary



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A Self-Transition

One time step still passes. Remaining in the same state is an outcome of choosing up.

The Components of an MDP

Symbol	Meaning	Robot example
$\mathcal{S}$	Possible states	Traversable grid cells
$\mathcal{A}(s)$	Available actions	Four directions
$T(s' s, a)$	Next-state probability	Intended move
$R(s, a, s')$	Immediate reward	Terminal entry or movement cost
γ	Discount factor	Weight placed on later rewards

- ▶ Specify the initial state or initial-state distribution.
- ▶ Specify when the task ends and how rewards accumulate.
- ▶ Here the transition and reward models are given to the agent.

States and Actions

$$\mathcal{S} = \{(x, y) : x \in \{1, 2, 3, 4\}, y \in \{1, 2, 3\}\} \setminus \{(2, 2)\}.$$

The model has eleven states, including the two terminal states.

Nonterminal States

$$\mathcal{A}(s) = \{\text{up, right, down, left}\}.$$

An action remains available even when its intended direction points into a wall or boundary.

Terminal States

$$s_+ = (4, 3), \quad s_- = (4, 2).$$

After terminal entry, the task is over.

The Markov Property

The current state must contain the information needed to predict the next state, given the current action:

$$P(S_{t+1} = s' \mid S_0, A_0, \dots, S_t = s, A_t = a) = T(s' \mid s, a).$$

- ▶ S_t is the state at time t ; A_t is the action taken there.
- ▶ Once S_t, A_t are known, earlier history adds no predictive information about S_{t+1} .
- ▶ Our reward is also determined by the current transition: $R(S_t, A_t, S_{t+1})$.

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A Modeling Requirement

The world does not need to forget its past. The state must summarize the relevant effects of that past.

Checkpoint: State Sufficiency

A Robot with a Battery

Movement success depends on the remaining battery charge. Two robots occupy the same cell, but one has a full battery and the other is nearly empty.

1. Do they have the same next-state distribution when selecting up?
2. Can position alone be a Markov state for this task?
3. What information should the state include?

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State Revision

Use $s = (x, y, b)$, where b is remaining battery charge, if these quantities determine the transition and reward model. The agent must observe all three components for this to be a fully observed MDP.

Transition Probabilities

For each state–action pair (s, a) ,

$$T(s' \mid s, a) \geq 0, \quad \sum_{s' \in \mathcal{S}} T(s' \mid s, a) = 1.$$

Grid Dynamics

- ▶ Move in the intended direction with probability 0.8.
- ▶ Move in each perpendicular direction with probability 0.1.
- ▶ If a resulting move hits a wall or boundary, remain in the current state.

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Grid Dynamics

- ▶ Move in the intended direction with probability 0.8.
- ▶ Move in each perpendicular direction with probability 0.1.
- ▶ If a resulting move hits a wall or boundary, remain in the current state.

Different movement outcomes can lead to the same next state. Add their probabilities when forming $T(s' \mid s, a)$.

Walls, Boundaries, and Self-Transitions

3	s			+1
2		wall		-1
1				
	1	2	3	4

Selected action: left at $s = (1, 3)$.

Movement	Next cell	Probability
Left	$(1, 3)$	0.8
Up	$(1, 3)$	0.1
Down	$(1, 2)$	0.1
Right	$(2, 3)$	0

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$$T((1, 3) \mid (1, 3), \text{left}) = 0.9,$$

$$T((1, 2) \mid (1, 3), \text{left}) = 0.1.$$

The blocked outcomes become self-transitions.

Rewards on Transitions

For a nonterminal current state s , our reward convention is

$$R(s, a, s') = \begin{cases} +1, & s' = s_+, \\ -1, & s' = s_-, \\ -0.04, & s' \text{ is nonterminal.} \end{cases}$$

- ▶ Receive the terminal reward when entering the terminal state.
- ▶ Every other transition costs 0.04, including staying after a blocked move.
- ▶ Terminal entry receives its listed reward alone; no extra movement cost is added.

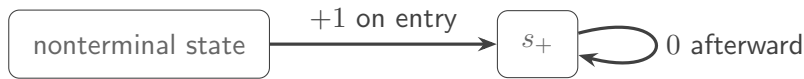
A Complete Convention

Specify when a reward is received. A number drawn in a cell does not settle its timing.

Terminal States and Absorbing Dynamics

For either terminal state $s \in \{s_+, s_-\}$,

$$T(s \mid s, \text{stay}) = 1, \quad R(s, \text{stay}, s) = 0.$$



- ▶ The episode ends at entry; the absorbing extension only adds zeros.
- ▶ The +1 or -1 is received once.
- ▶ Rewarding every terminal self-transition would define a different task.

Reading One State–Action Pair

At $s = (3, 2)$, the robot chooses right, toward the negative terminal.

Outcome	s'	$T(s' \mid s, \text{right})$	$R(s, \text{right}, s')$
Intended right	$(4, 2)$	0.8	-1
Up	$(3, 3)$	0.1	-0.04
Down	$(3, 1)$	0.1	-0.04
Left (blocked)	$(3, 2)$	0	-0.04

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Expected Immediate Reward

$$\sum_{s'} T(s' \mid s, \text{right}) R(s, \text{right}, s') = 0.8(-1) + 0.1(-0.04) + 0.1(-0.04) + 0(-0.04) = -0.808.$$

The expectation averages the possible outcomes.

Plans and Policies

A Fixed Action Sequence

up, up, right, right, right

From $(1, 1)$, these intended moves trace a route to s_+ .

A slip can leave the robot somewhere else while the sequence continues.

A Policy

$$\pi(s) \in \mathcal{A}(s)$$

A deterministic stationary policy assigns an action to every state.

After each move, observe the resulting state and choose $\pi(s)$ there.

Plans and Policies

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The Decision Depends on the Outcome

A policy specifies how to continue after intended moves, slips, and blocked moves.

A Policy

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A deterministic stationary policy assigns an action to every state.

After each move, observe the resulting state and choose $\pi(s)$ there.

A Policy for Every State

3	→	→	→	+1
2	↑	wall	↑	-1
1	↑	←	↑	←
	1	2	3	4

- ▶ Each arrow specifies an intended action.
- ▶ The transition model still determines the actual outcome.
- ▶ At $(1, 1)$, choose up.
- ▶ If the robot slips to $(2, 1)$, choose left on the next step.

The terminal action is stay. This is an illustrative policy; its optimality has not been established.

Policies and Sample Trajectories

A policy and the transition model generate a random sequence:

$$S_0, A_0, S_1, A_1, S_2, \dots, \quad A_t = \pi(S_t).$$

One possible successful trajectory under the illustrated policy is

$$(1, 1) \longrightarrow (1, 2) \longrightarrow (1, 3) \longrightarrow (2, 3) \longrightarrow (3, 3) \longrightarrow (4, 3).$$

Its rewards are

$$-0.04, \quad -0.04, \quad -0.04, \quad -0.04, \quad +1.$$

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One Realization

With no discounting, this trajectory returns $4(-0.04) + 1 = 0.84$. Repeating the same policy can produce different trajectories and returns.

Finite-Horizon Returns

If at most H actions remain, use $0 \leq \gamma \leq 1$ and define

$$G_0 = \sum_{t=0}^{H-1} \gamma^t R(S_t, A_t, S_{t+1}).$$

- ▶ H is the decision horizon: how many actions count.
- ▶ The first transition reward has weight 1.
- ▶ Later rewards have weights $\gamma, \gamma^2, \dots$
- ▶ After terminal entry, all subsequent rewards are zero.

Undiscounted Finite Horizon

Setting $\gamma = 1$ adds the rewards directly. A fixed finite horizon keeps the sum finite when rewards are bounded.

Discounting Later Rewards

Use $\gamma = 0.95$ for the five-transition trajectory:

$$G_0 = -0.04 - 0.95(0.04) - 0.95^2(0.04) - 0.95^3(0.04) + 0.95^4(1) \\ \approx 0.6661.$$

Smaller γ

Distant rewards have less influence. At $\gamma = 0$, only the next transition reward counts.

Larger γ

Distant rewards retain more influence. The horizon and terminal convention still matter.

Infinite-Horizon Returns

Without a fixed action limit, use

$$G_0 = \sum_{t=0}^{\infty} \gamma^t R(S_t, A_t, S_{t+1}), \quad 0 \leq \gamma < 1.$$

If $|R(s, a, s')| \leq R_{\max}$, then

$$|G_0| \leq \sum_{t=0}^{\infty} \gamma^t R_{\max} = \frac{R_{\max}}{1 - \gamma}.$$

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Why the Assumptions Matter

Bounded rewards and $\gamma < 1$ make the discounted sum well defined. An undiscounted infinite-horizon task needs additional assumptions; a robot that loops forever can accumulate an unbounded movement cost.

Policies With a Deadline

Remaining Time and Action Choice

An agent may prefer a nearby modest reward with one action left and a longer route to a larger reward with several actions left.

- ▶ A stationary policy uses only the current state: $\pi(s)$.
- ▶ A finite-horizon policy may also use the decision time: $\pi_t(s)$.
- ▶ Equivalently, include the remaining horizon in the state: (s, h) .

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- ▶ Equivalently, include the remaining horizon in the state: (s, h) .

Choose the Policy Class for the Task

For a fully observed MDP with finitely many states and actions, bounded rewards, and infinite horizon with $\gamma < 1$, an optimal deterministic stationary policy exists. A deadline can require time-dependent decisions.

Expected Return as the Objective

For a policy π and initial state s , write

$$J^\pi(s) = \mathbb{E}_\pi[G_0 \mid S_0 = s].$$

The expectation averages trajectories generated by the policy and transition model.

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Policy Objective

Choose a policy with the largest expected return:

$$\pi^* \in \arg \max_{\pi} J^\pi(s_0).$$

- ▶ A favorable sample trajectory does not establish that a policy is best.
- ▶ A good policy can still encounter an unfavorable random outcome.

Immediate Reward and Future Consequences

One-Step Evaluation

$$\mathbb{E}[R(S_0, A_0, S_1)]$$

This is the objective when only one action remains, or when $\gamma = 0$.

- ▶ Many moves in the grid have the same immediate reward of -0.04 .
- ▶ Their next states can have very different opportunities and risks.
- ▶ Choosing a locally attractive action can reduce total expected return.

Sequential Evaluation

$$\mathbb{E}_{\pi}[G_0]$$

This includes later rewards and the policy's future choices.

Rewards and the Task Objective

Model choice	What it expresses
Positive terminal reward	A desirable completion outcome
Negative terminal reward	An undesirable completion outcome
Negative movement reward	A cost of continuing to move
Discount factor	How later rewards are weighted
Decision horizon	Which future transitions count

Reward Design and Task Objectives

Changing the movement reward, terminal rewards, discount factor, or horizon can change the preferred policy even when transition probabilities stay fixed.

Checkpoint: Selecting Down Near the Goal

3			s	+1
2		wall		-1
1				
	1	2	3	4

At $s = (3, 3)$, choose down.

1. List each possible next state and its probability.
2. Assign a reward to each transition.
3. Compute the expected immediate reward.
4. If this action reaches s_+ , what reward follows on the absorbing transition?

Use the Complete Model

The intended move has probability 0.8; each perpendicular slip has probability 0.1. Terminal rewards are paid on entry.

Checkpoint Solution

Outcome	s'	$T(s' \mid (3, 3), \text{down})$	$R((3, 3), \text{down}, s')$
Down	$(3, 2)$	0.8	-0.04
Right	$(4, 3)$	0.1	+1
Left	$(2, 3)$	0.1	-0.04
Up (blocked)	$(3, 3)$	0	-0.04

$$\mathbb{E}[R \mid (3, 3), \text{down}] = 0.8(-0.04) + 0.1(1) + 0.1(-0.04) + 0(-0.04) = 0.064.$$

Checkpoint Solution

Outcome	s'	$T(s' \mid (3, 3), \text{down})$	$R((3, 3), \text{down}, s')$
Down	$(3, 2)$	0.8	-0.04
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Left	$(2, 3)$	0.1	-0.04
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$$\mathbb{E}[R \mid (3, 3), \text{down}] = 0.8(-0.04) + 0.1(1) + 0.1(-0.04) + 0(-0.04) = 0.064.$$

- ▶ Reaching s_+ gives +1 once; the following absorbing transition gives 0.
- ▶ The expected immediate reward is 0.064, even though no individual outcome gives that reward.
- ▶ This one-step calculation alone does not identify the best long-term policy.

Checkpoint: Model Repairs

Each claim describes a proposed robot model. What needs to change?

1. A wall collision removes that outcome's probability from the transition table.
2. The terminal cell pays $+1$ every time the robot stays there.
3. Battery charge affects motion, but the state records only position.
4. A task has a deadline, but the policy must choose the same action at a cell regardless of time remaining.

Model Repairs

1. Add the collision probability to the current state's self-transition so the distribution still sums to one.
2. Pay the terminal reward on entry and use zero reward on all later absorbing transitions.
3. Include the battery information needed to predict motion and rewards; ensure that the modeled state is observable.
4. Allow $\pi_t(s)$, or include remaining time in the state, so the policy can respond to the deadline.

Formulation Before Optimization

An optimization method answers the question encoded by the model. First make that question match the intended task.

MDP Formulation: Main Takeaways

- ▶ Choose a fully observed state that summarizes the history needed for prediction.
- ▶ Specify actions, normalized transition probabilities, and an explicit reward convention.
- ▶ Use a policy to choose actions after observing each resulting state.
- ▶ Define the horizon and discount factor, then maximize expected cumulative reward.

Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.